

MUHAMMAD SHAHRUKH KHAN

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PROFESSIONAL SUMMARY

Computer science student specializing in AI-powered applications, full-stack development, and RESTful API architecture with hands-on industry experience through multiple technical internships. Proven track record building production-grade systems using Python, FastAPI, Flutter, React.js, Docker, PostgreSQL, and Supabase. Experienced in machine learning, computer vision, and data science with a focus on real-world social impact. Proficient in Agile/Scrum workflows, Git version control, CI/CD pipelines, and end-to-end software development lifecycle (SDLC). Recognized for exceptional performance and rapid adaptation to emerging AI technologies.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C++, SQL, Assembly Language

Frameworks & Libraries: FastAPI, Flutter, React.js, NumPy, Pandas, Scikit-learn, TensorFlow

AI / ML Technologies: InsightFace, Computer Vision, Machine Learning, Generative AI, KDE, Cosine Similarity

Backend & Databases: RESTful APIs, PostgreSQL, Supabase, MongoDB, Database Design, SQL (SSMS)

DevOps & Tools: Docker, Git, GitHub, CI/CD Pipelines, Version Control, Postman

Methodologies: Agile/Scrum, SDLC, Object-Oriented Programming (OOP), Data Structures & Algorithms

EDUCATION

Bachelor of Science in Computer Science

Expected: May 2027

COMSATS University Islamabad, Wah Campus

Focus Areas: Artificial Intelligence · Software Engineering · Data Science · Cybersecurity

PROFESSIONAL EXPERIENCE

AI & ML Engineering Intern | DevelopersHub Corporation

June 2025 – July 2025

- Awarded Best Performance recognition for outstanding contributions across AI/ML engineering projects, ranking in the top 5% of intern cohort.
- Developed and deployed machine learning models using Python, Scikit-learn, NumPy, and Pandas — achieving up to 91% classification accuracy on structured datasets.
- Built and optimized RESTful API endpoints with FastAPI to serve ML model predictions, reducing average inference response time by ~30%.
- Applied data preprocessing, feature engineering, and model evaluation techniques (cross-validation, confusion matrix, ROC-AUC) following Agile sprint workflows.
- Collaborated with senior engineers via Git version control and participated in daily standups and code review sessions.

Python Programming Intern | Cosmicode Technologies

July 2025 – October 2025

- Completed 4-month internship contributing to backend development projects using Python, FastAPI, and PostgreSQL within a professional Agile/Scrum environment.
- Designed and implemented RESTful API modules with proper authentication, input validation, and error handling — improving system reliability and developer experience.
- Optimized SQL queries and database schemas in PostgreSQL, reducing query execution time by ~25% on high-frequency endpoints.
- Collaborated with cross-functional development teams on code reviews, unit testing, and CI/CD pipeline integration using Git and GitHub.

Python Programming Intern | CodeAlpha

June 2025

- Received Letter of Recommendation highlighting excellent analytical skills, rapid technology adoption, and consistent technical contributions.
- Developed Python-based applications including data processing scripts and automation tools using NumPy, Pandas, and standard library modules.
- Demonstrated strong problem-solving abilities in algorithm design and data structure implementation; fostered collaborative team environment through knowledge sharing.

KEY PROJECTS

SkillLoom – AI-Powered Textile Workforce Hiring Platform | Final Year Project

Tech Stack: *Python, FastAPI, Flutter, Supabase, PostgreSQL, Docker, InsightFace, React.js, Scikit-learn*

- Architected full-stack hiring platform targeting Pakistan's informal textile sector with 15M+ potential users, implementing complete SDLC from requirements engineering to deployment.
- Engineered AI-driven job matching system using cosine similarity for skills-based ranking and Haversine formula for location-based recommendations, improving match relevance by design.
- Implemented biometric identity verification using InsightFace (computer vision / facial recognition) to serve low-literacy worker populations — eliminating need for text-based login.
- Developed KDE-based demand heatmaps for real-time workforce planning and geographic labor market insights, enabling data-driven employer decisions.
- Built dual-interface system: Flutter mobile app (workers) and React.js admin portal (employers) with full CRUD operations, RESTful APIs, JWT authentication, and Docker containerization.

PMNH QR-Based Mobile Museum Guide | Commissioned Client Project

Tech Stack: *Python, FastAPI, Flutter, Supabase, PostgreSQL, RESTful APIs, Session Management*

- Delivered end-to-end backend system for Pakistan Museum of Natural History to digitize visitor experience through a mobile-first, WiFi-gated architecture with QR code scanning.
- Designed and deployed RESTful API backend handling exhibit content delivery, QR code resolution, session management, and visitor analytics with PostgreSQL data persistence.
- Built comprehensive admin portal for museum staff to manage exhibits, multimedia content, and visitor data — enabling non-technical staff to operate the system independently.
- Coordinated with client IT team to implement network-level access control and APK sideloading distribution strategy, completing Phase 1 MVP on schedule.

AI-Powered Bone Fracture Detection System

Tech Stack: *Python, Computer Vision, Machine Learning, Image Processing, Scikit-learn*

- Developed medical imaging application to detect and classify bone fractures from X-ray images using machine learning and image processing algorithms.
- Applied preprocessing, segmentation, and feature extraction techniques to determine fracture type and severity levels with measurable diagnostic accuracy.

Additional Projects

- University Management System (SQL/SSMS): Normalized relational database design managing student records, courses, attendance tracking, and grade computation.
- Library & Movie Ticket Systems (C++, OOP): Console applications implementing object-oriented design patterns, file I/O handling, and data structure algorithms.
- Python Portfolio: Location-based digital watch, MCQ scoring engine, and fitness tracking application demonstrating data structures and algorithm implementation.

CERTIFICATIONS

- **Generative AI Boot Camp** — Coursera, 2024
- **Python for Everybody (Python Fundamentals)** — Coursera / University of Michigan, 2024
- **Database Fundamentals (SQL & MongoDB)** — Coursera / IBM, 2024

LANGUAGES

English (Professional Fluency) • **Urdu** (Native)